Python Conditional Statements – Complete Notes

Introduction

Conditional statements are used to make decisions in a Python program. They allow the program to execute different blocks of code depending on whether a condition is True or False.

In real-world applications, conditional statements are used everywhere:

- E-commerce websites checking product availability
- Banking applications validating account balances
- Social media applications checking user authentication
- Online exam systems calculating grades
- Food delivery applications checking delivery availability

What is a Condition?

A condition is an expression that evaluates to either:

- True
- False

Example

```
temperature = 35

print(temperature > 30)
```

Output

```
True
```

The condition `temperature > 30` evaluates to True.

Comparison Operators

Comparison operators are used to compare values.

Operator	Meaning
==	Equal to
!=	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to

Example

```
price = 500

print(price == 500)
print(price > 300)
print(price < 200)
```

Output

```
True
True
False
```

if Statement

Definition

The if statement executes a block of code only when the condition is True.

Syntax

```
if condition:
   statements
```

Example

```
internet_connected = True

if internet_connected:
   print("Downloading file...")
```

Output

Downloading file...

Explanation

Since the condition is True, the code inside the if block executes.

Example: Product Availability

```
stock = 15

if stock > 0:
    print("Product available")
```

Output

Product available

if-else Statement

Definition

The `if-else` statement is used when there are two possible outcomes.

Syntax

```
if condition:
    statements
else:
    statements
```

Example

```
ticket_confirmed = False

if ticket_confirmed:
    print("Board the train")
else:
    print("Wait for confirmation")
```

Output

Wait for confirmation

Explanation

The condition is False, so the else block executes.

Example: Online Shopping

```
cart_total = 1200

if cart_total >= 1000:
    print("Free Delivery")
else:
    print("Delivery Charges Apply")
```

Output

Free Delivery

if-elif-else Statement

Definition

When multiple conditions need to be checked, use `if-elif-else`.

Syntax

```
if condition1:
    statements
elif condition2:
    statements
elif condition3:
    statements
else:
    statements
```

Example: Internet Speed Rating

```
speed = 75

if speed >= 100:
    print("Excellent")
elif speed >= 50:
    print("Good")
elif speed >= 20:
    print("Average")
else:
    print("Poor")
```

Output

```
Good
```

Example: Movie Ticket Pricing

```
age = 10

if age < 5:
    print("Free Entry")
elif age < 18:
    print("Child Ticket")
elif age < 60:
    print("Adult Ticket")
else:
    print("Senior Citizen Ticket")
```

Output

```
Child Ticket
```

Nested if Statement

Definition

An if statement inside another if statement is called a Nested if statement.

Syntax

```
if condition1:
    if condition2:
        statements
```

Example: Company Access System

```
employee = True
id_card = True

if employee:
    if id_card:
        print("Access Granted")
```

Output

```
Access Granted
```

Example: Online Course Enrollment

```
registered = True
fee_paid = True

if registered:
    if fee_paid:
        print("Course Activated")
```

Output

```
Course Activated
```

Logical Operators in Conditional Statements

Logical operators are used to combine multiple conditions.

AND Operator

Returns True only when all conditions are True.

Syntax

```
condition1 and condition2
```

Example

```
attendance = 85
project_submitted = True
```

```
if attendance >= 75 and project_submitted:
    print("Eligible for Certificate")
```

Output

```
Eligible for Certificate
```

OR Operator

Returns True if at least one condition is True.

Syntax

```
condition1 or condition2
```

Example

```
vip_member = False
premium_member = True

if vip_member or premium_member:
    print("Access Premium Content")
```

Output

```
Access Premium Content
```

NOT Operator

Reverses the result.

True becomes False and False becomes True.

Syntax

```
not condition
```

Example

```
maintenance_mode = False

if not maintenance_mode:
    print("Website Available")
```

Output

Website Available

Short Hand if

Used when only one statement needs to be executed.

Syntax

if condition: statement

Example

battery = 80

if battery > 50: print("Battery Healthy")

Output

Battery Healthy

Short Hand if-else (Ternary Operator)

Used to write if-else in a single line.

Syntax

value_if_true if condition else value_if_false

Example

temperature = 18

result = "Cold" if temperature < 20 else "Warm"

print(result)

Output

Cold

Multiple Conditions Using AND

Example

```
age = 30
income = 50000

if age >= 21 and income >= 30000:
    print("Credit Card Eligible")
```

Output

```
Credit Card Eligible
```

Multiple Conditions Using OR

Example

```
coupon = True
membership = False

if coupon or membership:
    print("Discount Applied")
```

Output

```
Discount Applied
```

Checking String Values

Conditional statements can compare strings.

Example

```
city = "Hyderabad"

if city == "Hyderabad":
    print("Welcome to Hyderabad")
```

Output

```
Welcome to Hyderabad
```

Checking Boolean Values

Example

```
notifications_enabled = True

if notifications_enabled:
    print("Sending Notifications")
```

Output

```
Sending Notifications
```

Using Membership Operators with Conditions

in Operator

Checks whether a value exists in a collection.

Example

```
languages = ["Python", "Java", "SQL"]

if "Python" in languages:
    print("Python Found")
```

Output

```
Python Found
```

not in Operator

Checks whether a value does not exist in a collection.

Example

```
languages = ["Python", "Java", "SQL"]

if "C++" not in languages:
    print("C++ Not Available")
```

Output

C++ Not Available

Real-Time Example: Food Delivery Application

```
restaurant_open = True
delivery_available = True

if restaurant_open and delivery_available:
    print("Order Accepted")
else:
    print("Service Unavailable")
```

Output

```
Order Accepted
```

Real-Time Example: OTT Subscription Check

```
subscription_active = True

if subscription_active:
    print("Enjoy Your Movie")
else:
    print("Please Renew Subscription")
```

Output

```
Enjoy Your Movie
```

Real-Time Example: Online Examination Result

```
score = 78

if score >= 90:
    print("Outstanding")
elif score >= 75:
    print("Very Good")
elif score >= 50:
    print("Good")
else:
    print("Needs Improvement")
```

Output

Very Good